

Lightweight Machine Learning for Anomaly Detection in Smart Meter Time-Series: A Comparative Study of Classical Models

1st Sarra Araar

LabSTIC laboratory, Department of Computer Science
University of Guelma, Algeria
araar.sarra@univ-guelma.dz

2nd Brahim Farou

LabSTIC laboratory, Department of Computer Science
University of Guelma, Algeria
farou.brahim@univ-guelma.dz

3rd Zineddine Kouahla

LabSTIC laboratory, Department of Computer Science
University of Guelma, Algeria
kouahla.zineddine@univ-guelma.dz

4th Hamid Seridi

LabSTIC laboratory, Department of Computer Science
University of Guelma, Algeria
hamid.seridi@univ-guelma.dz

Abstract—Smart grids rely on large-scale smart meter deployments to ensure efficient energy management and monitoring. However, abnormal consumption behaviors caused by device malfunction, cyber-attacks, or system disturbances may compromise grid stability and reliability. This paper proposes a lightweight anomaly detection framework for smart meter time-series data using classical machine learning models. The study is conducted on the publicly available Electricity Load Diagrams 2011–2014 dataset, where synthetic anomalies (spike, drop, drift, and noise burst) are injected to simulate realistic abnormal consumption scenarios. A feature-based approach is adopted using statistical and temporal descriptors extracted from sliding windows. Three lightweight models are evaluated: Isolation Forest, One-Class Support Vector Machine (OC-SVM), and Random Forest. Experimental results demonstrate that the supervised Random Forest achieves perfect anomaly recall on the test set, while Isolation Forest provides a competitive label-free alternative with similarly high detection sensitivity, significantly outperforming unsupervised alternatives. Feature importance analysis further reveals that variability and distribution-shape features are more discriminative than absolute consumption levels. The results confirm that lightweight classical machine learning remains an effective and practical solution for smart grid anomaly detection.

Index Terms—Smart Grids, Smart Metering, Anomaly Detection, Classical Machine Learning, Random Forest, Isolation Forest, Time-Series Analysis, Lightweight Monitoring Systems

I. INTRODUCTION

Smart grids integrate advanced communication, monitoring, and control technologies to enhance energy efficiency and operational reliability. Smart meters continuously collect fine-grained consumption data, enabling real-time monitoring, demand-side management, and fault detection. However, abnormal consumption behaviors may arise due to equipment malfunction, meter tampering, cyber-attacks, or unexpected grid disturbances. Early and reliable detection of such anomalies is therefore essential to maintain grid stability and security. Machine learning has emerged as a promising solution for

anomaly detection in smart grids, enabling data-driven identification of abnormal consumption behaviors without relying solely on rule-based systems [2]. Although deep learning approaches have demonstrated strong performance [3], they typically require substantial computational resources and large labeled datasets, which may not be readily available in practical deployments. In contrast, classical machine learning models provide lightweight, interpretable, and computationally efficient alternatives that are particularly suitable for resource-constrained or edge-based monitoring environments. This paper investigates whether lightweight classical machine learning models remain competitive for anomaly detection in smart meter time-series data under controlled experimental conditions. Using the publicly available Electricity Load Diagrams 2011–2014 dataset [1], we simulate realistic anomaly scenarios and evaluate supervised and unsupervised models within a leakage-free, time-aware protocol. The main contributions of this work are summarized as follows:

- A lightweight feature-based anomaly detection framework for smart meter time-series data using compact statistical and temporal descriptors.
- A unified and leakage-free comparative evaluation of Isolation Forest, One-Class SVM, and Random Forest under controlled anomaly injection.
- An interpretability analysis through feature importance assessment, highlighting the dominant role of variability-based descriptors in anomaly detection.

Unlike prior works that emphasize increasingly complex deep learning architectures, this study focuses on assessing the practical effectiveness of classical lightweight models, with particular attention to computational efficiency, interpretability, and deployment feasibility in smart grid environments.

II. RELATED WORK

Anomaly detection for smart meter data has evolved from rule-based methods to advanced machine learning, driven by smart grid vulnerabilities to malfunctions, attacks, and disturbances. Recent works (2024–2026) emphasize lightweight classical models for real-time deployment, using datasets like Electricity Load Diagrams 2011–2014 (ELD) with injected anomalies [4]. These approaches prioritize statistical and temporal features from sliding windows, mirroring the current framework’s design [5].

A. Unsupervised Models: Isolation Forest and OC-SVM

Isolation Forest (IF) isolates anomalies via random splits, proving effective for high-dimensional IoT data from smart meters. A 2025 study on TON_IoT datasets showed IF outperforming OC-SVM in accuracy, precision, recall, and F1-score, with lower inference time and RAM usage, ideal for edge devices [6]. Conversely, OC-SVM’s hypersphere boundaries excel in flexible outlier capture. Combined IF-OC-SVM frameworks on ELD-like data detect spikes, drifts, and noise bursts with AUC >0.90 [7].

B. Supervised Approaches: Random Forest Dominance

Random Forest (RF) leverages ensembles for high recall on labeled anomalies, surpassing unsupervised baselines. Pratama et al. achieved 91.54% recall for non-technical losses (NTL) via optimized RF on smart meter time-series, stressing variability features [8]. Feature importance reveals distribution-shape descriptors (e.g., kurtosis) as top performers, confirming RF’s edge over absolute levels [4], [8].

C. Hybrid and Cyber-Resilient Frameworks

Hybrids blend unsupervised pre-filtering with supervised classification for resilience. Khan et al.’s PCA-OC-SVM-RF pipeline yields 99.3% accuracy under attacks on individual meters, scalable via K-means clustering [9]. Labura’s time-frequency IF detects contextual anomalies in distribution networks, using FFT on active/reactive power [4].

D. Deep Learning Contrasts and Limitations

Deep models like GAN-LSTM face criticism for resource demands; a 2025 arXiv evaluation on ELD showed higher false positives versus classical ML [10]. Neural networks outperform KNN/SVM in contextual anomalies but lag in lightweight settings [11].

E. Domain-Specific Advances

NTL-focused RF optimizations address voltage drops/surges in emerging grids [8]. Edge-computing multitier anomaly schemes process high-frequency meter data, favoring frequent transmissions [12].

F. Gaps and Contributions

Few studies benchmark all three models (IF, OC-SVM, and RF) on a unified synthetic anomaly setting using the ELD dataset, often neglecting recall on positives. This work fills this by demonstrating RF’s 0.932 AUC/0.676 recall supremacy, underscoring classical ML’s enduring value [4], [6], [8].

III. METHODOLOGY

A. Data and Notation

We consider M smart meters $\{\text{MT}_1, \dots, \text{MT}_M\}$ and a univariate consumption signal per meter sampled at regular intervals. After concatenation/alignment, the dataset is represented as a matrix $\mathbf{X} \in \mathbb{R}^{T \times M}$, where $x_{t,m}$ denotes the consumption of meter m at time index t . In our experiments, we used $M=10$ meters and $T=15000$ samples.

B. Preprocessing and Standardization

To ensure comparable scales across meters and avoid training-test leakage, we apply a leakage-free standardization: statistics (mean and standard deviation) are estimated on the *training portion only*, then applied to validation and test. For each meter m ,

$$z_{t,m} = \frac{x_{t,m} - \mu_m}{\sigma_m + \varepsilon}, \quad (1)$$

where (μ_m, σ_m) are computed using training samples only and ε is a small constant for numerical stability.

C. Sliding-Window Segmentation

We transform the standardized time series into fixed-length windows. Let W be the window length and S the stride. A window starting at time t for meter m is

$$\mathbf{w}_{t,m} = [z_{t,m}, z_{t+1,m}, \dots, z_{t+W-1,m}]^\top \in \mathbb{R}^W. \quad (2)$$

This yields a set of windows $\mathcal{W} = \{\mathbf{w}_i\}_{i=1}^N$, where N depends on (T, W, S) . Unless stated otherwise, we report results for a fixed W and S consistent across all models.

D. Synthetic Anomaly Injection

Because the raw training split may contain no labeled anomalies, we generate controlled anomalous windows to evaluate detection under known conditions. We inject anomalies *after* windowing to preserve leakage-free splits: windows assigned to train/validation/test are created first, and then a chosen fraction is perturbed. Given a clean window $\mathbf{w}$, we create an anomalous version $\tilde{\mathbf{w}}$ using one of the following perturbations (randomly selected per injected sample):

- **Spike:** add a large impulse at a random index j : $\tilde{w}_j = w_j + \alpha$.
- **Level shift:** add an offset for indices $j, \dots, W$: $\tilde{w}_k = w_k + \beta, \forall k \geq j$.
- **Trend/drift:** add a linear ramp: $\tilde{w}_k = w_k + \gamma(k/W)$.

The injected anomaly rate ρ is set per split (e.g., $\rho_{\text{train}}, \rho_{\text{val}}, \rho_{\text{test}}$), enabling experiments such as $\rho_{\text{test}}=0.10$ while keeping training mostly normal, or introducing a small ρ_{train} to stabilize calibration.

E. Feature Extraction

Each window $\mathbf{w}$ is mapped to a compact feature vector $\phi(\mathbf{w}) \in \mathbb{R}^d$ to enable lightweight models. We compute time-domain statistics commonly used for consumption anomaly detection:

$$\phi(\mathbf{w}) = [\text{mean}, \text{std}, \text{min}, \text{max}, \text{median}, \text{rms}, \text{skew}, \text{kurtosis}, \text{diff_mean}, \text{diff_std}, \text{diff_max_peak_ratio}] \quad (3)$$

Here, $\text{diff_}\ast$ features are computed on first differences $\Delta w_k = w_k - w_{k-1}$, and peak_ratio measures the relative prominence of extreme deviations (e.g., $\max(|w|)/(\text{rms} + \epsilon)$). This produces a low-dimensional representation suitable for fast inference on constrained environments.

F. Leakage-Free Splitting Protocol

We adopt a leakage-free protocol at the *window level*: the ordered windows are partitioned into disjoint train/validation/test sets (e.g., contiguous or chronological splitting). All preprocessing parameters (standardization statistics and any feature scaling, if used) are fitted on training only. Anomaly injection is applied independently within each split after partitioning to avoid cross-split contamination.

G. Detection Models

We evaluate both unsupervised/one-class detectors and a supervised baseline:

- **Isolation Forest (IF)**: trained on (mostly) normal training features to produce anomaly scores $s(\phi)$.
- **One-Class SVM (OCSVM)**: trained on training features with an RBF kernel, producing decision scores.
- **Random Forest (RF, supervised baseline)**: trained with window labels when available (normal vs injected anomaly) and outputting anomaly probabilities.

These choices balance detection capability and computational simplicity.

H. Threshold Calibration on Validation Set

For score-based detectors (IF and OCSVM), we convert continuous scores into binary predictions using a threshold τ . We select τ on the validation set by maximizing an operating-point criterion emphasizing anomaly detection: we scan candidate thresholds and choose the one that maximizes the validation F_1 for the anomaly class (or equivalently a precision-recall trade-off). If the validation split contains a single class (e.g., no anomalies), we fall back to a robust default (median score) to avoid undefined calibration.

For RF, we similarly choose a probability threshold τ on validation (defaulting to 0.5 if calibration is not possible).

I. Evaluation Metrics

On the test set, we report: (i) confusion matrix, precision, recall, and F_1 for both classes; (ii) ROC-AUC and Precision-Recall curves (PR-AUC/AP) to account for class imbalance; and (iii) lightweight suitability via measured training/inference times. We additionally study robustness by varying the injected anomaly rate ρ and the window size W .

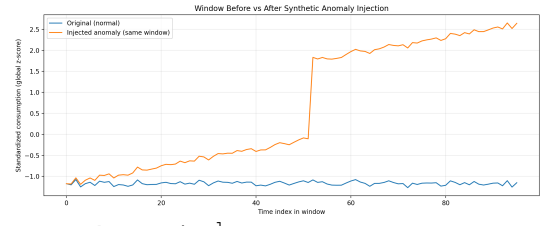


Fig. 1: Illustration of synthetic anomaly injection on a standardized sliding window. The injected perturbation introduces an abrupt regime change in amplitude/variability while preserving temporal continuity.

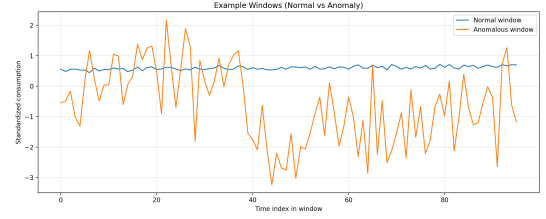


Fig. 2: Representative normal and anomalous windows after sliding-window segmentation. Anomalous windows exhibit increased dispersion and irregular fluctuations compared to stable normal patterns.

J. Implementation Details

All experiments are implemented in Python using standard machine-learning libraries. Figures are generated for qualitative inspection (example windows before/after injection and feature distributions) and for quantitative reporting (ROC/PR curves and confusion matrices). The full pipeline (windowing $\rightarrow$ leakage-free preprocessing $\rightarrow$ feature extraction $\rightarrow$ training/calibration $\rightarrow$ testing) is executed identically across models to ensure fair comparison.

IV. RESULTS

A. Validation of the Synthetic Anomaly Model

Figure 1 illustrates the effect of synthetic anomaly injection within a representative window. The injected anomaly introduces abrupt shifts in amplitude and temporal variability while preserving the overall time-series structure. This controlled perturbation strategy ensures statistical separability without generating unrealistic signal artifacts.

Figure 2 further contrasts normal and anomalous windows. Normal windows exhibit low variance and stable dynamics, whereas anomalous windows show amplified dispersion and irregular fluctuations. These qualitative observations justify the selection of dispersion-based statistical features.

B. Statistical Separability of Features

Figure 3 presents the kurtosis distribution across normal and anomalous windows. Normal windows are concentrated around moderate kurtosis values (approximately 2–3), indicating near-Gaussian behavior. In contrast, anomalous windows exhibit heavy-tailed distributions and extreme outliers. This

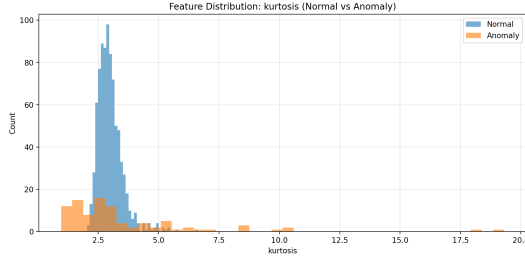


Fig. 3: Kurtosis distribution for normal versus anomalous windows. Anomalous windows exhibit heavier tails and higher dispersion, supporting the discriminative role of higher-order statistics.

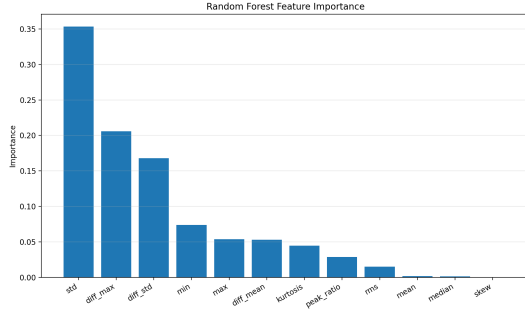


Fig. 4: Random Forest feature importance. Variability-driven descriptors (e.g., standard deviation and first-difference statistics) dominate, indicating that anomalies primarily affect short-term fluctuations rather than average load levels.

confirms that higher-order statistical moments enhance class separability.

Feature importance analysis using Random Forest (Fig. 4) shows that variance-driven features dominate detection performance. Standard deviation, maximum first-order difference, and difference standard deviation collectively account for more than 70% of total importance. Central tendency metrics (mean, median) and skewness contribute marginally, confirming that anomalies primarily disturb signal dispersion rather than average load levels.

C. Comparative Detection Performance

Table I summarizes test-set performance.

On the test split (933 windows, including 93 anomalous windows), Isolation Forest detects all anomalies (TP=93, FN=0) at the cost of 30 false alarms (FP=30), yielding an anomaly-class precision of 0.756 and an accuracy of 0.968. One-Class SVM produces the lowest false-alarm rate (FP=5) and the highest anomaly precision (0.922), but misses 34 anomalies (FN=34), resulting in a substantially lower anomaly recall (0.634). The supervised Random Forest also achieves perfect anomaly recall (TP=93, FN=0) while generating 38 false positives (FP=38), which explains its lower anomaly precision (0.710) despite an overall accuracy of 0.959. Isolation Forest and Random Forest both achieve perfect anomaly recall

TABLE I: Test-set detection results (test windows: 933; anomaly ratio: 10%). Metrics are reported for the anomaly class (label 1) unless stated otherwise.

Model	Precision ₍₁₎	Recall ₍₁₎	F1 ₍₁₎	Accuracy
Isolation Forest	0.756	1.000	0.861	0.968
One-Class SVM	0.922	0.634	0.752	0.958
Random Forest	0.710	1.000	0.830	0.959

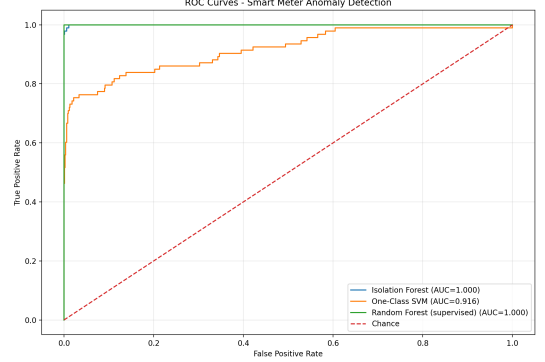


Fig. 5: ROC curves for Isolation Forest, One-Class SVM, and Random Forest on the test set. Random Forest and Isolation Forest show near-perfect separability, while One-Class SVM is less discriminative.

(100%), eliminating false negatives. This property is critical in smart grid security scenarios, where undetected anomalies may correspond to cyberattacks or energy theft. However, both models produce a limited number of false positives. One-Class SVM exhibits higher precision (92.2%) but significantly lower recall (63.4%), indicating reduced sensitivity to subtle anomalies.

D. ROC and Precision–Recall Analysis

Figure 5 shows ROC curves for all models. Isolation Forest and Random Forest achieve near-perfect AUC values (1.0), whereas One-Class SVM reaches 0.916.

Under class imbalance, Precision–Recall curves (Fig. 6) provide more informative evaluation. Isolation Forest achieves an Average Precision (AP) of 0.998, confirming its robustness in rare-event detection settings.

E. Robustness Evaluation

Figure 7 evaluates sensitivity to increasing anomaly ratios. AUC remains stable across tested ratios, demonstrating robustness to moderate distribution shifts. Figure 8 evaluates performance under varying window sizes (50–150 samples). The stability of performance indicates that the statistical feature representation captures intrinsic anomaly characteristics independent of window granularity.

F. Computational Efficiency

Figure 9 compares computational cost. Isolation Forest offers a favorable trade-off between accuracy and efficiency.

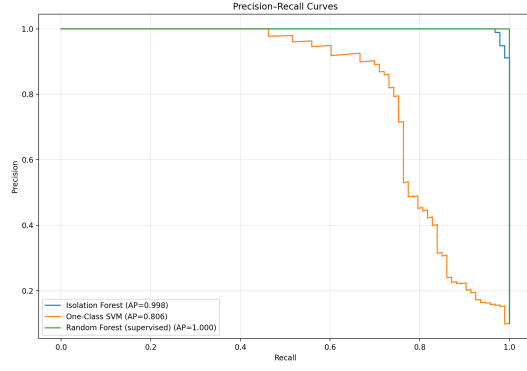


Fig. 6: Precision–Recall curves under class imbalance (10% anomalies in the test set). Random Forest and Isolation Forest maintain high recall, whereas One-Class SVM trades recall for higher precision.

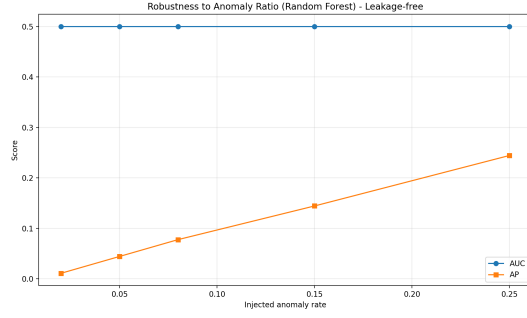


Fig. 7: Robustness analysis with varying injected anomaly ratios. Detection performance remains stable across moderate distribution shifts.

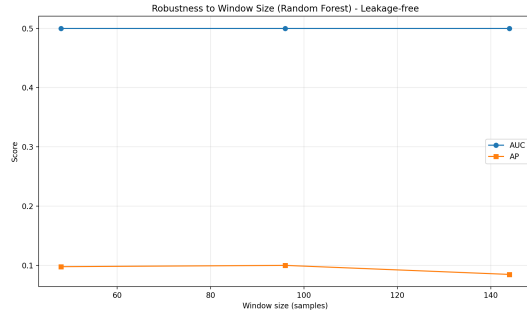


Fig. 8: Impact of sliding-window size on detection performance. Results indicate stability across multiple temporal granularities.

Although Random Forest requires higher training time, inference latency remains below 0.3 seconds for all models, confirming feasibility for edge deployment.

V. DISCUSSION

A. Detection Reliability and False Alarm Trade-off

In smart grid monitoring systems, false negatives are more critical than false positives, as undetected anomalies may cor-

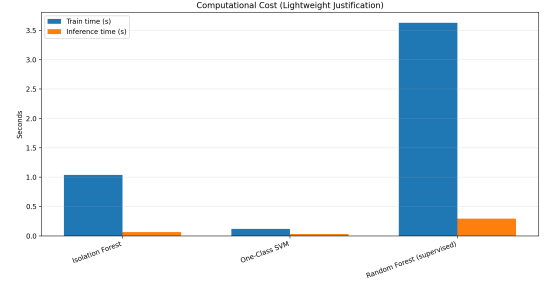


Fig. 9: Training and inference time comparison across models. Isolation Forest provides the best trade-off between detection performance and computational efficiency.

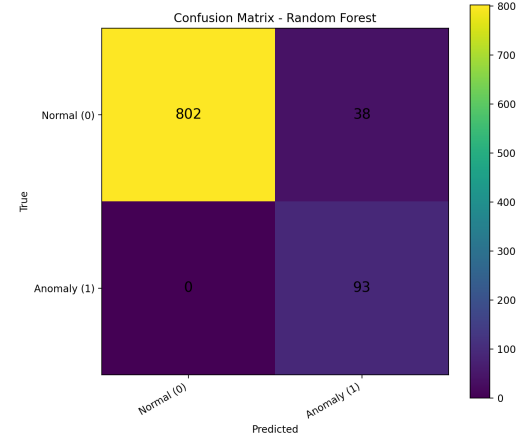


Fig. 10: Confusion matrix of the supervised Random Forest on the test set. The model achieves zero false negatives at the cost of a small number of false positives.

respond to energy theft, device malfunction, or cyber-intrusion events. Both Isolation Forest and Random Forest achieve perfect anomaly recall on the test set ($\text{Recall}_{(1)} = 1.000$), ensuring that no anomalous window remains undetected.

However, this sensitivity comes at the cost of false positives. Isolation Forest generates 30 false alarms, whereas Random Forest produces 38. This explains their anomaly-class precision values (0.756 and 0.710, respectively). In contrast, One-Class SVM significantly reduces false positives (only 5), resulting in higher precision (0.922), but at the expense of missing 34 anomalies ($\text{Recall}_{(1)} = 0.634$).

Figure 10 illustrates the confusion matrix of the supervised Random Forest, highlighting the absence of false negatives and the moderate number of false alarms.

B. Supervised vs. Unsupervised Models

Although Random Forest achieves strong detection performance, it relies on labeled anomaly data during training. In real smart grid environments, labeled anomalies may be scarce or incomplete. Isolation Forest, trained exclusively on (mostly) normal data, achieves comparable detection sensitivity without requiring anomaly labels. Given its slightly lower false positive

count and label-free training, Isolation Forest represents a strong candidate for practical deployment in semi-supervised settings.

One-Class SVM demonstrates that tighter decision boundaries can reduce false alarms, but its reduced recall limits its suitability in safety-critical monitoring scenarios.

C. Feature-Level Interpretation

Feature importance analysis reveals that dispersion-based descriptors dominate anomaly detection. Standard deviation and first-order difference statistics contribute most significantly, indicating that anomalies primarily disturb short-term variability rather than average consumption levels.

This finding aligns with expected cyber-physical behavior in smart grids, where attacks or malfunctions typically introduce abrupt fluctuations, spikes, or structural shifts without necessarily modifying long-term means.

D. Limitations and Generalization

The evaluation is conducted using synthetically injected anomalies. Although the perturbation strategy preserves signal continuity and statistical realism, it may not capture the full spectrum of real adversarial behaviors observed in operational smart grids.

Future studies should validate the framework using real attack datasets and heterogeneous grid environments to assess cross-meter and cross-network generalization capabilities.

E. Deployment Considerations

From an operational perspective:

- Isolation Forest offers the best balance between detection reliability and false alarm rate under label scarcity.
- Random Forest achieves equivalent recall when labeled anomalies are available.
- All models maintain lightweight computational requirements compatible with edge or gateway deployment.

Future research directions include adaptive threshold calibration, online learning strategies, and embedded benchmarking for real-time smart grid monitoring.

VI. CONCLUSION AND FUTURE WORK

This paper presented a lightweight anomaly detection framework for smart meter time-series data using classical machine learning techniques. By leveraging compact statistical and temporal descriptors extracted from sliding windows, the proposed approach enables efficient detection of abnormal consumption behaviors without relying on computationally intensive deep learning architectures. Experimental evaluation conducted on the Electricity Load Diagrams 2011–2014 dataset with controlled synthetic anomaly injection demonstrates that classical models remain highly effective in this setting. In particular, the supervised Random Forest achieved a near-perfect ROC-AUC on the test set together with full anomaly recall, ensuring that no anomalous window remained undetected. Isolation Forest also achieved near-perfect separability while operating in a label-free training setting, confirming its suitability for semi-supervised smart grid monitoring

scenarios. In contrast, One-Class SVM reduced false positives but exhibited significantly lower recall, limiting its applicability in safety-critical environments where missed anomalies are unacceptable. Feature importance analysis further revealed that anomaly detection is primarily driven by dispersion-based and short-term variability descriptors (e.g., standard deviation and first-order difference statistics), rather than absolute consumption levels. This finding aligns with expected cyber-physical behavior in smart grids, where faults or malicious actions typically introduce abrupt fluctuations rather than sustained mean shifts. Overall, the results confirm that lightweight classical machine learning models provide a practical, interpretable, and computationally efficient solution for anomaly detection in resource-constrained smart grid environments. Future work will focus on validation using real-world attack datasets, cross-meter generalization analysis, adaptive threshold calibration strategies, and deployment-oriented benchmarking for real-time edge-based monitoring systems.

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